PDQ: the week in numbers

4 October 2026. Nada.

	before	after
wrong answers (125 clips)	7	**1**
correct answers	52	48
mirrored clips identified	0 of 8	**8 of 8**
letterboxed clips identified	1 of 8	**6 of 8**

1. What was working before

The pod's PDQ engine, as an exact laptop copy (verified identical). Three parts can produce an answer, and they run in this order:

1. **The pre-scan.** Before anything else: 40 frames spread across the whole clip, a quick vote. If one film clearly owns those pictures, it answers and nothing else runs.

2. **The probe (P0).** If the pre-scan didn't answer: after about a second of reading the clip, one quick look - is any single frame close to some film? If yes, answer.

3. **The rules (P1-P9).** If neither quick part answered: keep reading, build a scorecard per candidate film, rules watch the scorecards, first to fire wins.

It genuinely was working, mostly: **52 of 125 known clips right, 7 wrong.**

2. How it was working, exactly

The pre-scan's vote

- a sampled frame is a "hit" on a film if its closest fingerprint is under distance 28
- a film is accepted on 3 hits with the best under 20
- never asks whether the hits agree where in the film they land, or whether they came from different parts of the clip

The probe's test

- first seconds only, one crop shape, up to 64 fingerprints, one search
- any single fingerprint under distance 30 -> answer. One frame, from the "probably" band, not the "certain" band

The rules

• the scorecard counts matched frames in 3 quality bands, keeps the best distance,

and keeps where in the film each hit lands (the offset) and which clip seconds produced it

• the careful rules use all of it - P4 wants 5 strong frames that agree on film

position AND come from 2 different moments of the clip

• but the rules only ever see the clips the quick parts passed over

3. Where the wrong answers came from

All 125 clips run, every answer tagged with the rule that made it:

part	wrong answers	the clips (brackets = distance)
pre-scan	4	Krrish 3 (6), The Northman (6), Civil War (12), The In Between (18)
probe P0	1	Smile (26.8, five frames)
rule P9E	1	Grave Encounters 2 (two frames, avg 34)
rule P8	1	Demeter (31 weak frames, avg 41)

```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_ORIGINAL.py

================================================================================
  WHICH RULE PRODUCED THE ANSWERS   (engine: pdq_engine_ORIGINAL.py)
================================================================================

  WRONG answers reproduced (these rules are the problem): 7
     SCAN       4
     P0         1
     P9E        1
     P8         1
       SCAN       Krrish 3                          avg=6.0 frames=27
       P0         Smile                             avg=26.8 frames=5
       SCAN       Captain America: Civil War        avg=12.0 frames=4
       P9E        Grave Encounters 2                avg=34.0 frames=2
       SCAN       The In Between                    avg=18.0 frames=4
       SCAN       The Northman                      avg=6.0 frames=6
       P8         The Last Voyage of the Demeter    avg=41.4 frames=31

  CORRECT answers (these rules must keep working): 52
     SCAN       46
     P0         3
     P3.5       2
     P1         1
```

• Distances 6 to 18 on the pre-scan four: those pictures genuinely are in our library,

attached to the wrong film (shared footage). The pre-scan answers before the rules run, so no rule change can ever touch them. The last three rules (P8, P9E, P0) are the ones I tightened in previous weeks.

So the actual problem:

• The engine knows how to check properly - it just doesn't check where it matters.

• The good checks exist: *do the matched frames point at the same place in the film?

did they come from more than one moment of the clip?* But they are written inside the mid-tier rules (P4, P6a, P7), and the rules only see a clip if the pre-scan and probe both passed on it first. In practice they almost never pass.

• So most clips get judged by the parts that ask the fewest questions, and the parts

that ask the right questions only see the leftovers.

• Four of the seven wrong answers came through the pre-scan, which never asked whether

the hits agreed on film position. They didn't - and that one question would have caught three of them.

4. What I built

1. Attribution testing - every clip run records which part of the engine gave the answer. Before, a wrong answer was just a number; now it has the name of the rule that made it next to it. The table above is its output.

2. The rebuilt probe. It samples 40 frames across the whole clip, like the pre-scan. For each sampled frame it fingerprints **every view**:

• A clip's frame is almost never the film frame as fingerprinted: the creator cropped

it for a phone screen, zoomed it, TikTok drew captions over it. The film is hiding somewhere inside the frame.

• So each frame is fingerprinted several ways: the whole frame, the caption-trimmed

frame, centre cut-outs in the common shapes at two zoom levels - and, after this week, the mirror of each and the view inside black bars.

• The old pre-scan never tried the caption-trimmed view; the old probe tried one shape

only. They could miss a match purely because of framing. The new probe makes every guess.

Per film it keeps a **scorecard**: distinct frames that hit it, best distance, and for every hit the pair (clip time, film time) so it can check the offsets. The old probe kept no scorecard at all - it decided on one number. This gives the deciding part the scorecard.

It answers in **three situations only**, each copied from a rule the pod already trusts. The ladder: the worse the pictures, the more agreement demanded.

bar	asks	copied from	impact (of its 41 answers)
PV-A	two frames basically identical to the film (under ~12-16)? One can be a fluke, two separate ones can't	P2.5	22 answers, e.g. mirrored Magadheera at distance 2
PV-B	a few decent frames (3 under 28, best under 20 - exactly the old pre-scan bar) AND two agreeing where in the film they sit? (that agreement is P4's check)	pre-scan + P4	17 answers; refuses Northman, Civil War, In Between
PV-C	only mediocre frames (5 under 33)? then prove it twice: 3 agreeing on film position, from 2 different clip moments	P6a	2 answers (heavily filtered clips)

And the old probe's question - "is there one frame under 30?" - **no longer exists**. That question is how Smile got wrongly answered; Smile now fails all three bars (best frame 22: not identical, not under 20) and the probe stays quiet.

5. The probe on real clips (real printouts, rerunnable)

A right answer - the Brick clip. 13 frames hit Brick, best distance 8, and three hits land at film minute 19.1 with the same offset. PV-B: decent frames, two agreeing

-> PASS. Answer: Brick.

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Brick clip> Brick
clip: vm_tiktok_com_ZN8rNvaKw_.mp4
film examined: 202981:Brick (tmdb=9270)

THE PROBE'S EVIDENCE for this film (13 sampled frames hit it):
  clip time   film time   distance      offset
    0.31min    97.10min          8    96.80min
    0.13min    42.59min         16    42.46min
    0.34min    88.16min         16    87.83min
    0.36min    67.31min         18    66.95min
    0.08min    19.14min         20    19.06min
    0.12min    19.17min         26    19.06min
    0.28min    17.12min         36    16.84min
    0.30min    74.77min         36    74.47min
    0.01min    29.46min         40    29.45min
    0.03min    19.09min         42    19.06min
    0.29min    17.15min         42    16.87min
    0.38min    45.03min         42    44.65min
    0.04min    19.09min         44    19.05min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=8 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=8 (needs <20)   frames under 28: 6 (needs 3)
        hits agreeing on one offset: 2 (needs 2)
        -> PASS

  PV-C  mediocre frames, proven twice over?
        best=8 (needs <26)   frames under 33: 6 (needs 5)
        agreeing: 2 (needs 3)   clip moments: 2 (needs 2)
        -> FAIL

VERDICT of the probe for this film: ANSWER: 202981:Brick (tmdb=9270)
```

The look-alike - the Northman clip. One frame at distance 6 (the shared footage), others at 16-44, but the offsets scatter: minutes 37, 40, 43, 49. No two agree. PV-A: only one near-perfect frame, fail. PV-B: decent count, agreement of one, fail. PV-C: fail. The probe stays quiet.

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Northman clip> Northman
clip: vm_tiktok_com_ZS4MtfV2y_This_post_is_shared_via_TikTok_Lite_Download_TikTok_Lite.mp4
film examined: 290:The Northman (tmdb=639933)

THE PROBE'S EVIDENCE for this film (6 sampled frames hit it):
  clip time   film time   distance      offset
    0.74min    49.62min          6    48.88min
    0.36min    40.31min         16    39.95min
    0.41min    43.39min         20    42.98min
    0.52min    49.13min         26    48.61min
    0.25min    36.96min         30    36.71min
    0.30min    37.02min         44    36.72min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=6 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=6 (needs <20)   frames under 28: 4 (needs 3)
        hits agreeing on one offset: 1 (needs 2)
        -> FAIL

  PV-C  mediocre frames, proven twice over?
        best=6 (needs <26)   frames under 33: 5 (needs 5)
        agreeing: 1 (needs 3)   clip moments: 5 (needs 2)
        -> FAIL

VERDICT of the probe for this film: STAYS QUIET - the rules (or nothing) take over
```

Both clips pass the old pre-scan's bar. On "count and best" they are indistinguishable - that is exactly how Northman got wrongly answered. The agreement question is the only thing that separates them.

The honest third case - the Soul clip. Five hits, all agreeing perfectly (every offset 32.2 minutes - a true scene's signature), but only two frames under 28 because the film occupies just the first seconds of the clip. Thin evidence, so the probe declines - and the fallback rules read the clip densely and name Soul via P1. Declined is not lost.

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Soul clip> Soul
clip: www_instagram_com_reel_DZ_L9fDykBP_.mp4
film examined: 265:Soul (tmdb=508442)

THE PROBE'S EVIDENCE for this film (5 sampled frames hit it):
  clip time   film time   distance      offset
    0.01min    32.24min          8    32.24min
    0.07min    32.29min         16    32.23min
    0.04min    32.24min         32    32.20min
    0.06min    32.27min         34    32.22min
    0.09min    32.31min         42    32.22min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=8 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=8 (needs <20)   frames under 28: 2 (needs 3)
        hits agreeing on one offset: 2 (needs 2)
        -> FAIL

  PV-C  mediocre frames, proven twice over?
        best=8 (needs <26)   frames under 33: 3 (needs 5)
        agreeing: 3 (needs 3)   clip moments: 1 (needs 2)
        -> FAIL

VERDICT of the probe for this film: STAYS QUIET - the rules (or nothing) take over
```

The two cases that remain, and why:

- **Krrish 3, the one wrong answer left.** Its pictures match at distance 6 AND agree

on film position - every test we have says genuine, because those exact frames really are in our library under Krrish 3 (reused footage). When pictures say yes in every way pictures can, only a different kind of evidence can say no: audio, SSCD.

- **The four lost correct answers - fan edits.** A fan edit cuts together moments of

one film, so its hits genuinely scatter - which is exactly what the Northman case looks like. Same evidence shape, opposite truth. No setting keeps the four and blocks the two, so it is a choice of which mistake to prefer. Per "a no answer beats a wrong one" I chose strict - that call is the policy question I'm bringing.

6. Results

engine	wrong	correct
pod's rules	7	52
new probe alone	1	41
new probe + rules backup	**1**	**48**

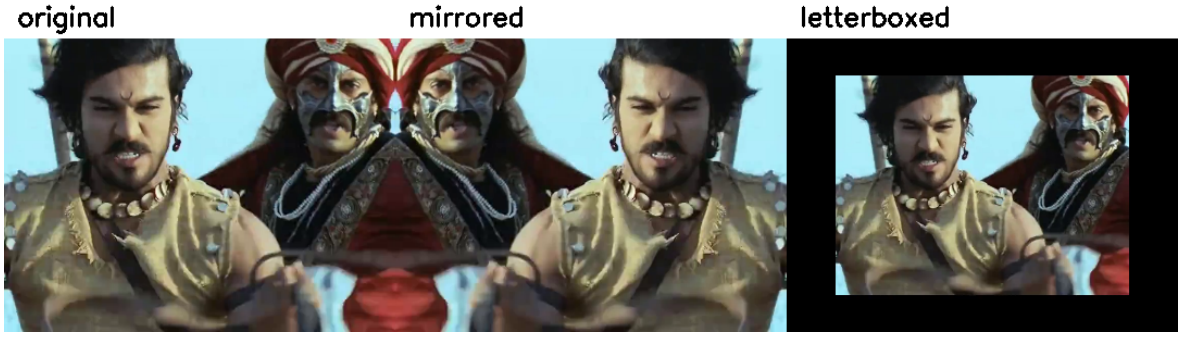
```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_PROBEV2.py

============================================================================
  WHICH RULE PRODUCED THE ANSWERS   (engine: pdq_engine_PROBEV2.py)
============================================================================

  WRONG answers reproduced (these rules are the problem): 1
     PV-A       1
       PV-A       Krrish 3                            avg=6.0 frames=16

  CORRECT answers (these rules must keep working): 48
     PV-A       22
     PV-B       17
     P1         4
     P3.5       3
     PV-C       2
```

Edited clips (10 reposter edits x 8 clips that work unedited):



edit	before	after
mirror	0 of 8	**8 of 8** (distance 2)
letterbox	1 of 8	**6 of 8**
filter / b&w / recompress / speeds	6-8 of 8	unchanged

- No edit ever produced a wrong film, on any version. Edits silence, never fool.
- Safety gate: all 125 real clips re-run with the fixes. Zero answers changed.

The week, before against after:

	before	after
wrong answers	7	**1** (Krrish 3: identical pictures, needs audio/SSCD)
correct answers	52	48 (the 4 lost are fan edits - the policy question)
agreement checked where most decisions happen	no	yes
minimum evidence for an answer	1 frame	2 near-identical frames
mirrored clips identified	0 of 8	**8 of 8**
letterboxed clips identified	1 of 8	**6 of 8**
wrong films caused by edits	0	0
every answer names its rule	no	yes
probe cost per clip	~30 s	~2 min (normal-views-first restores it)

The asks

1. Policy call: the 4 fan-edit clips. I recommend staying strict.
2. Someone watches the Misfits clip (both engines say F1, distance 8).
3. Direction: take the probe towards the pod? Next step: normal-views-first at pod scale.
4. Still open: keep clip_query_hashes rows for voted clips. Clips are still rotting.